



```
In [ ]: !pip install -q transformers accelerate pillow opencv-python
```

```
In [ ]: import getpass
api_key = getpass.getpass("please enter you gemini api key: ")
import google.generativeai as genai
genai.configure(api_key=api_key)
```

```
In [1]: !pip install -U "transformers==4.49.0" einops
```

Collecting transformers==4.49.0

Downloading transformers-4.49.0-py3-none-any.whl.metadata (44 kB)

44.0/44.0 kB 1.7 MB/s eta 0:00:00

Requirement already satisfied: einops in /usr/local/lib/python3.12/dist-packages (0.8.1)

Collecting einops

Downloading einops-0.8.2-py3-none-any.whl.metadata (13 kB)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers==4.49.0) (3.20.3)

Collecting huggingface-hub<1.0,>=0.26.0 (from transformers==4.49.0)

Downloading huggingface_hub-0.36.2-py3-none-any.whl.metadata (15 kB)

Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.49.0) (2.0.2)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.49.0) (25.0)

Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.49.0) (6.0.3)

Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.49.0) (2025.11.3)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from transformers==4.49.0) (2.32.4)

Collecting tokenizers<0.22,>=0.21 (from transformers==4.49.0)

Downloading tokenizers-0.21.4-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (6.7 kB)

Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.49.0) (0.7.0)

Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers==4.49.0) (4.67.1)

Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.26.0->transformers==4.49.0) (2025.3.0)

Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.26.0->transformers==4.49.0) (1.2.0)

Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.26.0->transformers==4.49.0) (4.15.0)

Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.49.0) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.49.0) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.49.0) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.49.0) (2026.1.4)

Downloading transformers-4.49.0-py3-none-any.whl (10.0 MB)

10.0/10.0 MB 62.5 MB/s eta 0:00:00

0:010:01

Downloading einops-0.8.2-py3-none-any.whl (65 kB)

65.6/65.6 kB 3.0 MB/s eta 0:00:00

Downloading huggingface_hub-0.36.2-py3-none-any.whl (566 kB)

566.4/566.4 kB 22.3 MB/s eta 0:00:00

0

Downloading tokenizers-0.21.4-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.8 MB)

6_64.whl (3.1 MB)

3.1/3.1 MB 56.7 MB/s eta 0:00:00:0

0:01

Installing collected packages: einops, huggingface-hub, tokenizers, transformers

Attempting uninstall: einops

Found existing installation: einops 0.8.1

Uninstalling einops-0.8.1:

Successfully uninstalled einops-0.8.1

Attempting uninstall: huggingface-hub

Found existing installation: huggingface_hub 1.4.1

Uninstalling huggingface_hub-1.4.1:

Successfully uninstalled huggingface_hub-1.4.1

Attempting uninstall: tokenizers

Found existing installation: tokenizers 0.22.2

Uninstalling tokenizers-0.22.2:

Successfully uninstalled tokenizers-0.22.2

Attempting uninstall: transformers

Found existing installation: transformers 5.2.0

Uninstalling transformers-5.2.0:

Successfully uninstalled transformers-5.2.0

Successfully installed einops-0.8.2 huggingface-hub-0.36.2 tokenizers-0.21.4 transformers-4.49.0

```
In [2]: from transformers import AutoProcessor, AutoModelForCausalLM
import torch

model_id = "microsoft/Florence-2-large"

processor = AutoProcessor.from_pretrained(model_id, trust_remote_code=True)
model = AutoModelForCausalLM.from_pretrained(
    model_id,
    torch_dtype=torch.float16,
    trust_remote_code=True
)

device = "cuda" if torch.cuda.is_available() else "cpu"
model.to(device)
model.eval()

print(f"loaded on {model.device}")
```

```

2026-03-06 08:16:57.778272: E external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to register cuFFT factory: Attempting to register factory for plugin cuFFT when one has already been registered
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
E0000 00:00:1772785018.025419      55 cuda_dnn.cc:8579] Unable to register cuDNN factory: Attempting to register factory for plugin cuDNN when one has already been registered
E0000 00:00:1772785018.091652      55 cuda_blas.cc:1407] Unable to register cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has already been registered
W0000 00:00:1772785018.628411      55 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1772785018.628462      55 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1772785018.628465      55 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.
W0000 00:00:1772785018.628467      55 computation_placer.cc:177] computation placer already registered. Please check linkage and avoid linking the same target more than once.

```

```

preprocessor_config.json:  0%|          | 0.00/806 [00:00<?, ?B/s]
processing_florence2.py: 0.00B [00:00, ?B/s]

```

A new version of the following files was downloaded from <https://huggingface.co/microsoft/Florence-2-large>:

- processing_florence2.py

. Make sure to double-check they do not contain any added malicious code. To avoid downloading new versions of the code file, you can pin a revision.

```

tokenizer_config.json:  0%|          | 0.00/34.0 [00:00<?, ?B/s]
vocab.json: 0.00B [00:00, ?B/s]
tokenizer.json: 0.00B [00:00, ?B/s]
config.json: 0.00B [00:00, ?B/s]
configuration_florence2.py: 0.00B [00:00, ?B/s]

```

A new version of the following files was downloaded from <https://huggingface.co/microsoft/Florence-2-large>:

- configuration_florence2.py

. Make sure to double-check they do not contain any added malicious code. To avoid downloading new versions of the code file, you can pin a revision.

```

modeling_florence2.py: 0.00B [00:00, ?B/s]

```

A new version of the following files was downloaded from <https://huggingface.co/microsoft/Florence-2-large>:

- modeling_florence2.py

. Make sure to double-check they do not contain any added malicious code. To avoid downloading new versions of the code file, you can pin a revision.

```

model.safetensors:  0%|          | 0.00/1.55G [00:00<?, ?B/s]
generation_config.json:  0%|          | 0.00/51.0 [00:00<?, ?B/s]
loaded on cuda:0

```

```

In [3]: import os
import json

```

```

page_id = "page_001"

base_dir = f"outputs/{page_id}"
lines_dir = f"{base_dir}/lines"

os.makedirs(lines_dir, exist_ok=True)

```

```

In [4]: from PIL import Image

image_path = "/kaggle/input/datasets/shubhamshukla02/testdata/page1.png"

image = Image.open(image_path).convert("RGB")

image.save(f"{base_dir}/main_image.png")

```

```

In [5]: import os

image_path = "/kaggle/input/datasets/shubhamshukla02/testdata/page1.png"

page_id = os.path.splitext(os.path.basename(image_path))[0]

```

```

In [6]: model = AutoModelForCausalLM.from_pretrained(
        model_id,
        torch_dtype=torch.float16,
        trust_remote_code=True
    ).to("cuda").eval()

```

```

In [7]: import torch
import cv2
import os
import json
import numpy as np
from PIL import Image

image_path = "/kaggle/input/datasets/shubhamshukla02/testdata/page1.png"
page_id = "page1"

base_dir = f"outputs/{page_id}"
lines_dir = f"{base_dir}/lines"
os.makedirs(lines_dir, exist_ok=True)
image = Image.open(image_path).convert("RGB")
image.save(f"{base_dir}/main_image.png")
task_prompt = "<MORE_DETAILED_CAPTION>"

print(f" Florence-2...")
inputs = processor(text=task_prompt, images=image, return_tensors="pt").to("cu

with torch.no_grad():
    generated_ids = model.generate(
        input_ids=inputs["input_ids"],
        pixel_values=inputs["pixel_values"],

```

```

        max_new_tokens=512,
        num_beams=3,
        do_sample=False
    )

    generated_text = processor.batch_decode(generated_ids, skip_special_tokens=False)
    parsed_answer = processor.post_process_generation(
        generated_text,
        task=task_prompt,
        image_size=(image.width, image.height)
    )

    metadata_text = parsed_answer[task_prompt]
    print(f"ok metadata: {metadata_text[:100]}...")
    img_cv = cv2.imread(image_path)
    img_gray = cv2.cvtColor(img_cv, cv2.COLOR_BGR2GRAY)
    thresh = cv2.adaptiveThreshold(
        img_gray, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
        cv2.THRESH_BINARY_INV, 35, 15
    )
    projection = np.sum(thresh, axis=1)
    threshold_value = (img_gray.shape[1] * 255) * 0.05

    lines = []
    start = None

    for i, val in enumerate(projection):
        if val > threshold_value and start is None:
            start = i
        elif val <= threshold_value and start is not None:
            if (i - start) > 10:
                lines.append((start, i))
            start = None
    line_paths = []
    for idx, (y1, y2) in enumerate(lines):
        pad = 2
        y1_pad, y2_pad = max(0, y1-pad), min(img_gray.shape[0], y2+pad)

        crop = img_cv[y1_pad:y2_pad, :]
        line_filename = f"line_{idx:02d}.png"
        line_path = os.path.join(lines_dir, line_filename)
        cv2.imwrite(line_path, crop)
        line_paths.append(line_path)

    metadata = {
        "page_id": page_id,
        "source_image": image_path,
        "description": metadata_text,
        "total_lines_found": len(line_paths),
        "lines": line_paths
    }

    with open(f"{base_dir}/metadata.json", "w") as f:

```

```

        json.dump(metadata, f, indent=2)

print(f"{len(line_paths)} lines. Data saved to {base_dir}/metadata.json")

```

Florence-2...

ok metadata: The image is a photograph of an old, handwritten document. The document appears to be old and worn, ...

16 lines. Data saved to outputs/page1/metadata.json

```

In [8]: import torch
import os
from transformers import Qwen2VLForConditionalGeneration, AutoProcessor
from peft import PeftModel, PeftConfig
from safetensors.torch import load_file
model_id = "Qwen/Qwen2-VL-2B-Instruct"
adapter_path = "/kaggle/input/datasets/shubhamshukla02/qwen-finetuned/qwen_rea
model = Qwen2VLForConditionalGeneration.from_pretrained(
    model_id,
    torch_dtype=torch.float16,
    device_map="auto"
)

print("qwen done")
config = PeftConfig.from_pretrained(adapter_path)
model = PeftModel(model, config)
adapter_file = os.path.join(adapter_path, "adapter_model.safetensors")
state_dict = load_file(adapter_file)
new_state_dict = {}
for key, value in state_dict.items():
    new_key = key.replace("base_model.model.model", "base_model.model")
    new_state_dict[new_key] = value

model.load_state_dict(new_state_dict, strict=False)

try:
    processor = AutoProcessor.from_pretrained(adapter_path, trust_remote_code=
except Exception as e:
    print(f"issue: {e}")
    processor = AutoProcessor.from_pretrained(model_id, trust_remote_code=True

model.eval()
print("-" * 30)
print("ok")
print("-" * 30)

```

config.json: 0.00B [00:00, ?B/s]

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Downloading shards: 0%| | 0/2 [00:00<?, ?it/s]

model-00001-of-00002.safetensors: 0%| | 0.00/3.99G [00:00<?, ?B/s]

model-00002-of-00002.safetensors: 0%| | 0.00/429M [00:00<?, ?B/s]

Loading checkpoint shards: 0%| | 0/2 [00:00<?, ?it/s]

generation_config.json: 0%| | 0.00/272 [00:00<?, ?B/s]

qwen done

issue: /kaggle/input/datasets/shubhamshukla02/qwen-finetuned/qwen_read_lora does not appear to have a file named preprocessor_config.json. Checkout 'https://huggingface.co//kaggle/input/datasets/shubhamshukla02/qwen-finetuned/qwen_read_lora/tree/main' for available files.

preprocessor_config.json: 0%| | 0.00/347 [00:00<?, ?B/s]

Using a slow image processor as `use\_fast` is unset and a slow processor was saved with this model. `use\_fast=True` will be the default behavior in v4.48, even if the model was saved with a slow processor. This will result in minor differences in outputs. You'll still be able to use a slow processor with `use\_fast=False`.

tokenizer_config.json: 0.00B [00:00, ?B/s]

vocab.json: 0.00B [00:00, ?B/s]

merges.txt: 0.00B [00:00, ?B/s]

tokenizer.json: 0.00B [00:00, ?B/s]

chat_template.json: 0.00B [00:00, ?B/s]

ok

WE LOADED FINE TUNED QWEN

In [9]: **import** json

metadata_path = "outputs/page1/metadata.json"

with open(metadata_path) **as** f:
 metadata = json.load(f)

line_paths = metadata["lines"]

print("lines", len(line_paths))

lines 16

In [10]: **from** PIL **import** Image, ImageOps
import torch

results = []

for line_path **in** line_paths:
 image = Image.open(line_path).convert("RGB")
 w, h = image.size
 pad_w = max(0, 28 - w)
 pad_h = max(0, 28 - h)

if pad_w > 0 **or** pad_h > 0:
 image = ImageOps.expand(image, (0, 0, pad_w, pad_h), fill="white")
 messages = [
 {
 "role": "user",
 "content": [
 {"type": "image"},


```

        {"type": "text", "text": "Transcribe the handwritten text."}
    ]
}
]

text_prompt = processor.apply_chat_template(
    messages,
    tokenize=False,
    add_generation_prompt=True
)
inputs = processor(
    images=image,
    text=text_prompt,
    return_tensors="pt"
).to(model.device)

with torch.no_grad():
    output = model.generate(
        **inputs,
        max_new_tokens=100,
        do_sample=False
    )
prediction = processor.decode(
    output[0][len(inputs.input_ids[0]):],
    skip_special_tokens=True
)

results.append({
    "line": line_path,
    "text": prediction
})

print(line_path, "→", prediction)

```

```

/usr/local/lib/python3.12/dist-packages/transformers/generation/configuration_utils.py:629: UserWarning: `do_sample` is set to `False`. However, `temperature` is set to `0.01` -- this flag is only used in sample-based generation modes. You should set `do_sample=True` or unset `temperature`.

```

```

warnings.warn(

```

```

/usr/local/lib/python3.12/dist-packages/transformers/generation/configuration_utils.py:634: UserWarning: `do_sample` is set to `False`. However, `top_p` is set to `0.001` -- this flag is only used in sample-based generation modes. You should set `do_sample=True` or unset `top_p`.

```

```

warnings.warn(

```

```

/usr/local/lib/python3.12/dist-packages/transformers/generation/configuration_utils.py:651: UserWarning: `do_sample` is set to `False`. However, `top_k` is set to `1` -- this flag is only used in sample-based generation modes. You should set `do_sample=True` or unset `top_k`.

```

```

warnings.warn(

```

outputs/pagel/lines/line_00.png → The handwritten text in the image is:

```

Información de afiliación hidalúrgica y minera de Sáxerre  
unida a la Diputación de esta U. de Algeciras  
alcalde del señor Alcalde de esta U. de Algeciras  
y alcalde de la Diputación de esta U. de Algeciras  
y alcalde de la Diputación de esta U. de Al

outputs/pagel/lines/line\_01.png → The handwritten text is:

"des de la configuración."

outputs/pagel/lines/line\_02.png → The handwritten text is:

"Undar de Muquiza nial y deino deita Villa."

outputs/pagel/lines/line\_03.png → Anuc Um pouco Comomas dia Suax; Hidio que

outputs/pagel/lines/line\_04.png → The handwritten text is:

"Soy hijo de Domingo de Muñoz y Álvarez."

outputs/pagel/lines/line\_05.png → The handwritten text is:

"de sauracowina na Difuntos házido iprocreado"

outputs/pagel/lines/line\_06.png → el matrimonio de los de Welles, Hijo por part  
e de la duena de Juan de Miguazza y la cabana de Sa

outputs/pagel/lines/line\_07.png → The handwritten text is:

"Haizicouli Loxma mufers aporla Mactina de"

outputs/pagel/lines/line\_08.png → The handwritten text is:

"Baix de Sauracouinita y Tonha de Canaregu"

This translates to:

"Lower of Sauracouinita and Tonha of Canaregu"

outputs/pagel/lines/line\_09.png → manida y mula Lomos todos Verinos deuadha

outputs/pagel/lines/line\_10.png → Villa quinta y Yo romos Hobbe hijos daleos de  
dan

outputs/pagel/lines/line\_11.png → The handwritten text in Spanish is as follow  
s:

```plaintext

que Christianos Cifos sin vara alguna de Moxor
Judios Moqués, Penitenciado por la Santa
Inquisición ni de otra Secta Nacuada, yantes Vien
dezendientes Léoninos alo Primero Poblado
us dierta M.H. Y.M.S. Provincia de Guipuz

```

outputs/pagel/lines/line\_12.png → The handwritten text is:

"coa yonimarios y dependientes de las Casas Pila"

outputs/pagel/lines/line\_13.png → The handwritten text is:

"13 Infanzones de Muxuruza Beyna, Suva y noca"

outputs/pagel/lines/line\_14.png → The handwritten text is:

"Xia ennta Villa dela de Sagartecueta Tita en  
la Villa de Meyba, dela de Gallarizegui, Tita  
de la Villa de Meyba, dela de Gallarizegui, Tita"  
outputs/pagel/lines/line\_15.png → The handwritten text is:

"En la Villa de Bejara y el de Sanxacujo Vitta"

Actual Result Predicted my our fine tuned Qwen

```
In [11]: import json
import os

clean_results = []

for res in results:
 raw_text = res['text']

 if ":" in raw_text:
 actual_prediction = raw_text.split(":")[-1].strip()
 else:
 actual_prediction = raw_text.strip()

 actual_prediction = actual_prediction.replace('`', '').replace("'", '').

 clean_results.append({
 "line": res['line'],
 "transcription": actual_prediction
 })

final_para = " ".join([item['transcription'] for item in clean_results if item

with open(f"{base_dir}/fine_tuned_output.json", "w") as f:
 json.dump(clean_results, f, indent=2)

with open(f"{base_dir}/actual_manuscript_text.txt", "w") as f:
 f.write(final_para)

print("qwen(fine tuned)output")
print(final_para)
print(f"\n fine tuned output{base_dir}/actual_manuscript_text.txt")
```

```
qwen(fine tuned)output
Información de afiliación hidalúrgica y minera de Sáxerre
unida a la Diputación de esta U. de Algeciras
alcalde del señor Alcalde de esta U. de Algeciras
y alcalde de la Diputación de esta U. de Algeciras
y alcalde de la Diputación de esta U. de Al des de la configuración. Undar de M
uquiza nial y deino deita Villa. Anuc Um pouco Comomas dia Suax; Hidio que Soy
hijo de Domingo de Muñoz y Álvarez. de sauracowina na Difuntos házido iprocread
o el matrimonio de los de Welles, Hijo por parte de la duena de Juan de Miguazz
a y la cabana de Sa Haizicouli Loxma mufers aporla Mactina de Lower of Sauracou
inita and Tonha of Canaregu manida y mula Lomos todos Verinos deuadha Villa qui
nta y Yo romos Hobbe hijos daleos dedan plaintext
que Christianos Cifos sin vara alguna de Moxor
Judios Moqués, Penitenciado por la Santa
Inquisición ni de otra Secta Nacuada, yantes Vien
dezendientes Léoninos alo Primero Poblado
us dierta M.H. Y.M.S. Provincia de Guipuz coa yonimarios y dependientes de las
Casas Pila 13 Infanzones de Muxuruza Beyna, Suva y noca Xia ennta Villa dela de
Sagartecueta Tita en
la Villa de Meyba, dela de Gallarizegui, Tita
de la Villa de Meyba, dela de Gallarizegui, Tita En la Villa de Bejara y el de
Sanxacujo Vitta
```

```
fine tuned outputoutputs/pagel/actual_manuscript_text.txt
```

```
In [12]: pip install -U google-genai
```

```
huggingface/tokenizers: The current process just got forked, after parallelism
has already been used. Disabling parallelism to avoid deadlocks...
To disable this warning, you can either:
 - Avoid using `tokenizers` before the fork if possible
 - Explicitly set the environment variable TOKENIZERS_PARALLELISM=(true
| false)
```

Requirement already satisfied: google-genai in /usr/local/lib/python3.12/dist-packages (1.55.0)  
Collecting google-genai  
 Downloading google\_genai-1.66.0-py3-none-any.whl.metadata (51 kB)  
 

---

52.0/52.0 kB 1.9 MB/s eta 0:00:00

Requirement already satisfied: anyio<5.0.0,>=4.8.0 in /usr/local/lib/python3.12/dist-packages (from google-genai) (4.12.1)  
Collecting google-auth<3.0.0,>=2.47.0 (from google-auth[requests]<3.0.0,>=2.47.0->google-genai)  
 Downloading google\_auth-2.48.0-py3-none-any.whl.metadata (6.2 kB)  
Requirement already satisfied: httpx<1.0.0,>=0.28.1 in /usr/local/lib/python3.12/dist-packages (from google-genai) (0.28.1)  
Requirement already satisfied: pydantic<3.0.0,>=2.9.0 in /usr/local/lib/python3.12/dist-packages (from google-genai) (2.12.3)  
Requirement already satisfied: requests<3.0.0,>=2.28.1 in /usr/local/lib/python3.12/dist-packages (from google-genai) (2.32.4)  
Requirement already satisfied: tenacity<9.2.0,>=8.2.3 in /usr/local/lib/python3.12/dist-packages (from google-genai) (9.1.2)  
Requirement already satisfied: websockets<17.0,>=13.0.0 in /usr/local/lib/python3.12/dist-packages (from google-genai) (15.0.1)  
Requirement already satisfied: typing-extensions<5.0.0,>=4.11.0 in /usr/local/lib/python3.12/dist-packages (from google-genai) (4.15.0)  
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from google-genai) (1.9.0)  
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (from google-genai) (1.3.1)  
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5.0.0,>=4.8.0->google-genai) (3.11)  
Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from google-auth<3.0.0,>=2.47.0->google-auth[requests]<3.0.0,>=2.47.0->google-genai) (0.4.2)  
Requirement already satisfied: cryptography>=38.0.3 in /usr/local/lib/python3.12/dist-packages (from google-auth<3.0.0,>=2.47.0->google-auth[requests]<3.0.0,>=2.47.0->google-genai) (43.0.3)  
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.12/dist-packages (from google-auth<3.0.0,>=2.47.0->google-auth[requests]<3.0.0,>=2.47.0->google-genai) (4.9.1)  
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0,>=0.28.1->google-genai) (2026.1.4)  
Requirement already satisfied: httpcore==1.\* in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0,>=0.28.1->google-genai) (1.0.9)  
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.\*->httpx<1.0.0,>=0.28.1->google-genai) (0.16.0)  
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.9.0->google-genai) (0.7.0)  
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.9.0->google-genai) (2.41.4)  
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.9.0->google-genai) (0.4.2)  
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.28.1->google-genai) (3.4.4)  
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.28.1->google-genai) (2.5.0)  
Requirement already satisfied: cffi>=1.12 in /usr/local/lib/python3.12/dist-packages

kages (from cryptography>=38.0.3->google-auth<3.0.0,>=2.47.0->google-auth[requests]<3.0.0,>=2.47.0->google-genai) (2.0.0)  
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /usr/local/lib/python3.12/dist-packages (from pyasn1-modules>=0.2.1->google-auth<3.0.0,>=2.47.0->google-auth[requests]<3.0.0,>=2.47.0->google-genai) (0.6.2)  
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=1.12->cryptography>=38.0.3->google-auth<3.0.0,>=2.47.0->google-auth[requests]<3.0.0,>=2.47.0->google-genai) (2.23)  
Downloading google\_genai-1.66.0-py3-none-any.whl (732 kB)

732.2/732.2 kB 15.4 MB/s eta 0:00:00  
000:01

Downloading google\_auth-2.48.0-py3-none-any.whl (236 kB)  
236.5/236.5 kB 16.4 MB/s eta 0:00:00

0  
Installing collected packages: google-auth, google-genai

Attempting uninstall: google-auth  
Found existing installation: google-auth 2.43.0  
Uninstalling google-auth-2.43.0:  
Successfully uninstalled google-auth-2.43.0

Attempting uninstall: google-genai  
Found existing installation: google-genai 1.55.0  
Uninstalling google-genai-1.55.0:  
Successfully uninstalled google-genai-1.55.0

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

bigframes 2.31.0 requires google-cloud-bigquery-storage<3.0.0,>=2.30.0, which is not installed.

google-adk 1.21.0 requires google-cloud-bigquery-storage>=2.0.0, which is not installed.

google-colab 1.0.0 requires google-auth==2.43.0, but you have google-auth 2.48.0 which is incompatible.

google-colab 1.0.0 requires jupyter-server==2.14.0, but you have jupyter-server 2.12.5 which is incompatible.

google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.3.3 which is incompatible.

Successfully installed google-auth-2.48.0 google-genai-1.66.0

Note: you may need to restart the kernel to use updated packages.

In [13]: **import** cv2

```
img = cv2.imread("/kaggle/input/datasets/shubhamshukla02/testdata/page1.png")
print(img.shape)
```

(942, 645, 3)

In [14]: **import** cv2

```
img = cv2.imread("/kaggle/input/datasets/shubhamshukla02/testdata/page1.png")

scale = 2
high_res = cv2.resize(img, None, fx=scale, fy=scale, interpolation=cv2.INTER_C
cv2.imwrite("page_300dpi.jpg", high_res)
```

Out[14]: True

```
In [15]: gray = cv2.cvtColor(high_res, cv2.COLOR_BGR2GRAY)
cv2.imwrite("page_gray.jpg", gray)
```

Out[15]: True

```
In [16]: pip install scikit-image
```

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS\_PARALLELISM=(true | false)

Requirement already satisfied: scikit-image in /usr/local/lib/python3.12/dist-packages (0.25.2)

Requirement already satisfied: numpy>=1.24 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (2.0.2)

Requirement already satisfied: scipy>=1.11.4 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (1.16.3)

Requirement already satisfied: networkx>=3.0 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (3.6.1)

Requirement already satisfied: pillow>=10.1 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (11.3.0)

Requirement already satisfied: imageio!=2.35.0,>=2.33 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (2.37.2)

Requirement already satisfied: tifffile>=2022.8.12 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (2026.1.14)

Requirement already satisfied: packaging>=21 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (25.0)

Requirement already satisfied: lazy-loader>=0.4 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (0.4)

Note: you may need to restart the kernel to use updated packages.

```
In [17]: from skimage.filters import threshold_sauvola
import numpy as np

window_size = 25
thresh = threshold_sauvola(gray, window_size=window_size)

binary = gray > thresh
binary = (binary * 255).astype("uint8")

cv2.imwrite("page_sauvola.jpg", binary)
```

Out[17]: True

```
In [18]: pip install scikit-image
```

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS\_PARALLELISM=(true | false)

Requirement already satisfied: scikit-image in /usr/local/lib/python3.12/dist-packages (0.25.2)

Requirement already satisfied: numpy>=1.24 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (2.0.2)

Requirement already satisfied: scipy>=1.11.4 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (1.16.3)

Requirement already satisfied: networkx>=3.0 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (3.6.1)

Requirement already satisfied: pillow>=10.1 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (11.3.0)

Requirement already satisfied: imageio!=2.35.0,>=2.33 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (2.37.2)

Requirement already satisfied: tifffile>=2022.8.12 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (2026.1.14)

Requirement already satisfied: packaging>=21 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (25.0)

Requirement already satisfied: lazy-loader>=0.4 in /usr/local/lib/python3.12/dist-packages (from scikit-image) (0.4)

Note: you may need to restart the kernel to use updated packages.

In [19]: `from skimage.morphology import skeletonize`

```
skeleton = skeletonize(binary // 255)
skeleton = (skeleton * 255).astype("uint8")

cv2.imwrite("page_skeleton.jpg", skeleton)
```

Out[19]: True

In [20]: `import google.generativeai as genai`  
`from PIL import Image`

```
genai.configure(api_key=api_key)

model = genai.GenerativeModel("gemini-2.5-flash")

image = Image.open("page_sauvola.jpg")

prompt = """
You are a 16th-century Spanish paleography expert.

Transcribe the handwritten manuscript exactly.

Rules:
- Maintain original spelling
- Do not modernize words
- If strokes overlap, infer using grammatical context
```



```
- Preserve line breaks if possible
"""

response = model.generate_content([prompt, image])

print(response.text)
```

/usr/local/lib/python3.12/dist-packages/wrapt/importer.py:223: FutureWarning:

All support for the `google.generativeai` package has ended. It will no longer be receiving updates or bug fixes. Please switch to the `google.genai` package as soon as possible.

See README for more details:

<https://github.com/google-gemini/deprecated-generative-ai-python/blob/main/README.md>

```
self.__wrapped__.exec_module(module)
```

Aquí está la transcripción del manuscrito, manteniendo la ortografía original y la estructura de las líneas:

ynformacion defiuacion Hidalguia p[ro]moüra de Baxen  
con la annee.l Señor Alcalde de neüra V[ieja] y Algonax opocion  
q[ue] Andres de Muguruza y Saca de sus Gracias y  
los y Confirmacion.

Andres de Muguruza natural y Bezino de esta Villa  
Anue vm parezco Comomas aia Lugar; Y digo q[ue]  
Soy hijo Lexmo de Domingo de Muguruza y Anna  
de Sagarrecoinia ya Difuntos Hazido y procraedo  
en el Matrimonio Lexmo de ellos, Hizo por parte.  
Laurena de Juan de Muguruza y Catalina de Ba-  
larrzauri su Lexma muxer y por la Mariena de  
Barr: de Sagarrecoinia y Josepha de Lanizagui  
marido y muxer Lexmos todos Vecinos de esta  
Villa quienes y lo somos Nobles hijos dalgo limpios  
y Christianos Viejos sin Vara alguna de Moros  
Judios ni Morizcos, Penitenciados por la Santa In-  
quisicion ni de otra Secta Reprouada, y ante vm  
Duzendú más Lexmos de los Primeros Poblado-  
res de esta M.N.Y.M.L. Prouinzia de Guipuz-  
coa y originarios y dependientes de las Casas Sola-  
res y Infanzonas de Muguruza Reyna Suyá y noua  
ria en esta Villa de la Sagarreconia Sita en  
la Villa de Alegubax, de la de Gallarzui, Sita  
en la Villa de Beroizaga y de la de Lanuzeguill Sita

THE OUTPUT FROM THE GEMINI 2.5-FLASH (replace the api key with urs)

```
In [21]: with open("gemini_raw.txt","w") as f:
 f.write(response.text)
```

```
In [22]: with open("gemini_raw.txt","r") as f:
```

```
raw_vlm_output = f.read()
```

```
In [23]: import PIL.Image
from google import genai

client = genai.Client(api_key=api_key)

img = PIL.Image.open('/kaggle/input/datasets/shubhamshukla02/testdata/page1.pr

with open("gemini_raw.txt", "r", encoding="utf-8") as f:
 gemini_output = f.read()

qwen_path = "outputs/page1/actual_manuscript_text.txt"
with open(qwen_path, "r", encoding="utf-8") as f:
 qwen_output = f.read()

final_judge_prompt = f"""
You are an expert editor of early modern Spanish manuscripts.
Compare these two candidate transcriptions against the provided IMAGE.

Candidate 1 (Gemini):
{gemini_output}

Candidate 2 (Qwen):
{qwen_output}

Task: Use the image to resolve differences.
Rules:
1. DO NOT modernize spelling.
2. Preserve historical orthography (ç, v for u, f for h).
3. Preserve line structure.
4. Output ONLY the final corrected text.
"""

response = client.models.generate_content(
 model="gemini-flash-latest",
 contents=[img, final_judge_prompt]
)

print(response.text)
```

Informacion defiliacion hidalguía y Limpieza de Sangre  
unida ante el Señor Alcalde deesta Va d Elgoibar a pedimto  
d Andres de Muguruza y Saca departidas de Baupito Casa-  
dos y Confirmacion ==

Andres de Muguruza nral y Vezino deesta Villa  
Ante Vm parezco Comomas aia Lugar; Y digo que  
Soy hijo Lexmo de Domingo de Muguruza y Antonia  
de Sagarrecomina ya Difuntos: Hazido y procreado  
enel Matrimonio Lexmo deellos, Nieto por parte  
Paterna de Juan de Muguruza y Catalina de Ba-  
llairteguí su Lexma muser y porla Materna de  
Barthme de Sagarrecomina y Josepha de Larrategui  
marido y muser Lexmos todos Vezinos deesta dha  
Villa quienes y Yo somos Nobles hijos dalgo de San-  
gre Christianos Viejos sin Raza alguna de Moros  
Judios: Hagonéss, Penitenciados porla Santa In-  
quisicion ni de otra Secta Reprouada, y antes Bien  
deszendiéntes Lexittimos delos Primeros Poblado-  
res deesta M.N. Y M.L. Prouinzia de Guipuz-  
coa y originarios y dependientes delas Casas Sola-  
res Ymfanzonas de Muguruza Reyna Sita y notta-  
ria enesta Villa dela de Sagarrecomina Sita en  
la Villa de Aleybar, dela de Gallaittegui, Sita  
enla Villa de Bergara y dela de Larrategui sitta

In [ ]: THE FINAL LLM OUTPUT FROM THE (GEMINI OUTPUT+IMAGE+QWEN OUTPUT)

```
In [24]: output_file_path = "final_answer"

with open(output_file_path, "w", encoding="utf-8") as f:
 f.write(response.text)

print(f"output saved: {output_file_path}")
```

output saved: final\_answer

```
In [25]: with open("/kaggle/working/final_answer.txt", "w", encoding="utf-8") as f:
 f.write(response.text)

import os
if os.path.exists("/kaggle/working/final_answer.txt"):
 print("saved")
else:
 print("prblm")
```

saved

```
In [28]: import re
import Levenshtein

def get_clean_string(text):
 return re.sub(r'\s+', '', text)

try:
 with open("/kaggle/working/final_answer.txt", "r", encoding="latin-1") as
```

```

 pred_raw = f.read()

 with open("/kaggle/input/datasets/shubhamshukla02/ground-truth/ground_truth.txt", "r", encoding="latin-1") as f:
 gt_raw = f.read()

 pred_clean = get_clean_string(pred_raw)
 gt_clean = get_clean_string(gt_raw)

 distance = Levenshtein.distance(gt_clean, pred_clean)
 cer = (distance / len(gt_clean)) * 100

 print(f"errors: {distance}")
 print(f"cer score: {cer:.2f}%")

except FileNotFoundError as e:
 print(f"error: {e}")

```

errors: 86  
cer score: 8.92%

```

In [32]: import re
from evaluate import load
bertscore = load("bertscore")
try:
 with open("/kaggle/working/final_answer.txt", "r", encoding="latin-1") as f:
 pred_raw = f.read()
 with open("/kaggle/input/datasets/shubhamshukla02/ground-truth/ground_truth.txt", "r", encoding="latin-1") as f:
 gt_raw = f.read()
 predictions = [pred_raw]
 references = [gt_raw]
 results = bertscore.compute(predictions=predictions, references=references, lang="en", model_type="distilbert-base-uncased")

 print(f"F1 score: {results['f1'][0]:.4f}")
 print(f"Precision score: {results['precision'][0]:.4f}")
 print(f"Recall score: {results['recall'][0]:.4f}")

except FileNotFoundError as e:
 print(f"error is there: {e}")

```

F1 score: 0.9107  
Precision score: 0.9080  
Recall score: 0.9134

```

In []: import re
import Levenshtein

def get_clean_words(text):
 text = text.lower()
 text = re.sub(r'^\w\s', '', text)
 return text.split()

try:
 with open("/kaggle/working/final_answer.txt", "r", encoding="latin-1") as f:

```

```

 pred_raw = f.read()

 with open("/kaggle/input/datasets/shubhamshukla02/ground-truth/ground_truth.txt") as f:
 gt_raw = f.read()
 pred_words = get_clean_words(pred_raw)
 gt_words = get_clean_words(gt_raw)
 distance = Levenshtein.distance(gt_words, pred_words)
 wer_score = (distance / len(gt_words)) * 100
 print(f"word error: {distance}")
 print(f"GT all words: {len(gt_words)}")
 print(f"wer score: {wer_score:.2f}%")

except FileNotFoundError as e:
 print(f"err: {e}")

```

In [30]: !pip install evaluate bert-score

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...  
 To disable this warning, you can either:

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS\_PARALLELISM=(true | false)

Collecting evaluate

Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)

Collecting bert-score

Downloading bert\_score-0.3.13-py3-none-any.whl.metadata (15 kB)

Requirement already satisfied: datasets>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from evaluate) (4.0.0)

Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from evaluate) (2.0.2)

Requirement already satisfied: dill in /usr/local/lib/python3.12/dist-packages (from evaluate) (0.3.8)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from evaluate) (2.3.3)

Requirement already satisfied: requests>=2.19.0 in /usr/local/lib/python3.12/dist-packages (from evaluate) (2.32.4)

Requirement already satisfied: tqdm>=4.62.1 in /usr/local/lib/python3.12/dist-packages (from evaluate) (4.67.1)

Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from evaluate) (3.6.0)

Requirement already satisfied: multiprocessing in /usr/local/lib/python3.12/dist-packages (from evaluate) (0.70.16)

Requirement already satisfied: fsspec>=2021.05.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]>=2021.05.0->evaluate) (2025.3.0)

Requirement already satisfied: huggingface-hub>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from evaluate) (0.36.2)

Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from evaluate) (25.0)

Requirement already satisfied: torch>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from bert-score) (2.9.0+cu126)

Requirement already satisfied: transformers>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from bert-score) (4.49.0)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from bert-score) (3.10.0)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets>=2.0.0->evaluate) (3.20.3)

Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.0.0->evaluate) (18.1.0)

Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.0.0->evaluate) (6.0.3)

Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]>=2021.05.0->evaluate) (3.13.3)

Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.7.0->evaluate) (1.2.0)

Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.7.0->evaluate) (4.15.0)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->evaluate) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->evaluate) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->evaluate) (2025.3)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->evaluate) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->evaluate) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->evaluate) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->evaluate) (2026.1.4)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (75.2.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (3.6.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (3.1.6)

Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (12.6.77)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (12.6.77)

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (12.6.80)

Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (9.10.2.21)

Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (12.6.4.1)

Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (11.3.0.4)

Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (10.3.7.77)

Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (11.7.1.2)

Requirement already satisfied: nvidia-cuspars-cu12==12.5.4.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (12.5.4.2)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (0.7.1)

Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (2.27.5)

Requirement already satisfied: nvidia-nvshmem-cu12==3.3.20 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (3.3.20)

Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (12.6.77)

Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (12.6.85)

Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (1.11.1.6)

Requirement already satisfied: triton==3.5.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.0.0->bert-score) (3.5.0)

Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers>=3.0.0->bert-score) (2025.11.3)

Requirement already satisfied: tokenizers<0.22,>=0.21 in /usr/local/lib/python3.12/dist-packages (from transformers>=3.0.0->bert-score) (0.21.4)

Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers>=3.0.0->bert-score) (0.7.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->bert-score) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->bert-score) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->bert-score) (4.61.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->bert-score) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib->bert-score) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->bert-score) (3.3.1)

Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (2.6.1)

Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (1.4.0)

Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (25.4.0)

Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (1.8.0)

Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (6.7.0)

Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (0.4.1)

Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (1.22.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->evaluate) (1.17.0)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.0.0->bert-score) (1.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.0.0->bert-score) (3.0.3)

Downloading evaluate-0.4.6-py3-none-any.whl (84 kB)

84.1/84.1 kB 2.1 MB/s eta 0:00:00a

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Downloading bert\_score-0.3.13-py3-none-any.whl (61 kB)

61.1/61.1 kB 4.5 MB/s eta 0:00:00

Installing collected packages: evaluate, bert-score

Successfully installed bert-score-0.3.13 evaluate-0.4.6

In [27]: !pip install python-Levenshtein

huggingface/tokenizers: The current process just got forked, after parallelism has already been used. Disabling parallelism to avoid deadlocks...

To disable this warning, you can either:

- Avoid using `tokenizers` before the fork if possible
- Explicitly set the environment variable TOKENIZERS\_PARALLELISM=(true | false)



Collecting python-Levenshtein

Downloading python\_levenshtein-0.27.3-py3-none-any.whl.metadata (3.9 kB)

Collecting Levenshtein==0.27.3 (from python-Levenshtein)

Downloading levenshtein-0.27.3-cp312-cp312-manylinux\_2\_24\_x86\_64.manylinux\_2\_28\_x86\_64.whl.metadata (3.7 kB)

Collecting rapidfuzz<4.0.0,>=3.9.0 (from Levenshtein==0.27.3->python-Levenshtein)

Downloading rapidfuzz-3.14.3-cp312-cp312-manylinux\_2\_27\_x86\_64.manylinux\_2\_28\_x86\_64.whl.metadata (12 kB)

Downloading python\_levenshtein-0.27.3-py3-none-any.whl (9.5 kB)

Downloading levenshtein-0.27.3-cp312-cp312-manylinux\_2\_24\_x86\_64.manylinux\_2\_28\_x86\_64.whl (153 kB)

153.3/153.3 kB 4.4 MB/s eta 0:00:00

Downloading rapidfuzz-3.14.3-cp312-cp312-manylinux\_2\_27\_x86\_64.manylinux\_2\_28\_x86\_64.whl (3.2 MB)

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Installing collected packages: rapidfuzz, Levenshtein, python-Levenshtein

Successfully installed Levenshtein-0.27.3 python-Levenshtein-0.27.3 rapidfuzz-3.14.3